nature machine intelligence

Article

https://doi.org/10.1038/s42256-023-00734-1

Construction of a 3D whole organism spatial atlas by joint modelling of multiple slices with deep neural networks

Received: 16 February 2023

Accepted: 11 September 2023

Published online: 19 October 2023

Check for updates

Gefei Wang [1,8], Jia Zhao [1,8], Yan Yan [2,3], Yang Wang [1,4], Angela Ruohao Wu [3,5,6,7] & Can Yang [1,4]

Spatial transcriptomics (ST) technologies are revolutionizing the way to explore the spatial architecture of tissues. Currently, ST data analysis is often restricted to a single two-dimensional (2D) tissue slice, limiting our capacity to understand biological processes that take place in 3D space. Here we present STitch3D, a unified framework that integrates multiple ST slices to reconstruct 3D cellular structures. By jointly modelling multiple slices and integrating them with single-cell RNA-sequencing data, STitch3D simultaneously identifies 3D spatial regions with coherent gene-expression levels and reveals 3D cell-type distributions. STitch3D distinguishes biological variation among slices from batch effects, and effectively borrows information across slices to assemble powerful 3D models. Through comprehensive experiments, we demonstrate STitch3D's performance in building comprehensive 3D architectures, which allow 3D analysis in the entire tissue region or even the whole organism. The outputs of STitch3D can be used for multiple downstream tasks, enabling a comprehensive understanding of biological systems.

Spatial transcriptomics (ST) technologies enable high-throughput gene expressions profiling in intact tissues[1–4]. Recently, datasets containing multiple parallel two-dimensional (2D) slices in tissues have been accumulated[5–8], but most existing tools only provide analysis for one single slice. Organs are organized as complex structures in 3D space, and biological processes rarely occur in a single 2D plane. For example, morphogen gradients spread in 3D space to instruct cell-type differentiation and morphogenesis to organize into functional organs during embryogenesis[9]. Having only a 2D view greatly restricts our interpretation of biological signalling and processes, how they influence the organization of different cell types, and how cells interact with each other.

With each single slice characterizing a 2D transcriptomic landscape, joint modelling of multiple slices offers opportunities to depict 3D pictures of biological systems[10]. To this end, additional efforts with joint analysis of multiple slices to reconstruct detailed 3D tissue structures are needed. Specifically, the following two 3D analysis tasks are fundamental. The first is to identify biologically interpretable 3D spatial regions with spots having similar gene expressions for revealing tissue structures. The result then enables downstream analyses like detection of region-related genes with 3D spatial patterns. The second task is to infer 3D fine-grained cell-type distributions by integrating multiple ST slices and single-cell RNA-sequencing (scRNA-seq) atlases.

[1]Department of Mathematics, The Hong Kong University of Science and Technology, Hong Kong SAR, China. [2]Shenzhen PKU-HKUST Medical Center, Shenzhen, China. [3]Division of Life Science, The Hong Kong University of Science and Technology, Hong Kong SAR, China. [4]Guangdong-Hong Kong-Macao Joint Laboratory for Data-Driven Fluid Mechanics and Engineering Applications, The Hong Kong University of Science and Technology, Hong Kong SAR, China. [5]Department of Chemical and Biological Engineering, The Hong Kong University of Science and Technology, Hong Kong SAR, China. [6]State Key Laboratory of Molecular Neuroscience, The Hong Kong University of Science and Technology, Hong Kong SAR, China. [7]Center for Aging Science, The Hong Kong University of Science and Technology, Hong Kong SAR, China. [8]These authors contributed equally: Gefei Wang, Jia Zhao. e-mail: angelawu@ust.hk; macyang@ust.hk

Existing next-generation sequencing ST technologies can detect transcriptome-wide gene expressions within spatial spots, but each spot often contains multiple cells, resulting in a relatively lower resolution[11]. Utilizing information from scRNA-seq atlases, the 3D cell-type deconvolution task decomposes cell-type mixtures in spatial spots and thereby achieves higher resolution for 3D reconstruction, allowing insight into biological functions of specific cell-type-enriched areas.

In this Article we present STitch3D for characterizing 3D tissue architectures using multiple slices. It addresses the aforementioned two 3D analysis tasks in a unified framework to provide complementary views of 3D tissue architectures. By effectively modelling gene expressions and spatial locations from multiple slices, STitch3D can distinguish biological variation among slices from batch effects, and integrate information across slices to assemble powerful 3D tissue models. Benchmarking studies and real data examples illustrate STitch3D's state-of-the-art accuracy and ability to borrow information across multiple slices. Compared to existing ST tools, most of which are developed for analysing one single slice by performing either the spatial domain detection task[12–14] or the cell-type deconvolution task[15–19], STitch3D enables 3D views of cellular tissue architectures with accurate 3D domain detection and 3D cell-type deconvolution results. Using diverse datasets, we demonstrate STitch3D's 3D reconstruction capability at the tissue scale or even for a whole organism. STitch3D's outputs can be applied to various downstream tasks, showing its versatility. STitch3D is a publicly available package (https://github.com/YangLabHKUST/STitch3D), and serves as an efficient and reliable tool for ST studies.

Results

Method overview

STitch3D is a deep learning-based method that uses multiple 2D ST slices to reconstruct 3D tissue structures (Fig. 1). The inputs of STitch3D are multiple ST slices and a matched scRNA-seq reference. Preprocessing steps include building 3D spatial spot coordinates by aligning slices, and constructing the global 3D neighbourhood graph. After these steps, STitch3D is trained to integrate information across all slices. A shared latent space is introduced to distil meaningful biological variations and facilitate the removal of batch effects. In the latent space, each spot has its representation, which is used to perform spatial domain identification and cell-type deconvolution tasks jointly.

Specifically, STitch3D maps gene expressions and spatial information of spots from multiple slices into the shared latent space by a graph attention-based network that leverages the 3D neighbourhood graph for spots. Another neural network is introduced to infer cell-type proportions from latent representations. With effective batch-effect modelling for integrating multiple slices, STitch3D is trained to reconstruct ST gene expressions by combining estimated cell-type proportions with cell-type-specific gene-expression profiles. After training, STitch3D learns spatially informed spot representations and cell-type proportions. The representations are used for 3D spatial domain identification with clustering algorithms, while the cell-type proportions help recover 3D cell-type distributions. With these outputs, STitch3D enables various downstream analyses. Details are included in the Methods.

Benchmarking studies

We first evaluated STitch3D's spatial domain detection performance using a human dorsolateral prefrontal cortex (DLPFC) dataset[6]. Six layers (L1–L6) and white matter (WM) were annotated in the original study. When applied to each single slice, STitch3D stably recovered the layer structures. When applied to multislice analysis, STitch3D yielded consistent results across slices, facilitating 3D reconstruction (Supplementary Fig. 1). STitch3D's result has a similar pattern compared to layer annotations, indicating its reliability (Fig. 2a,b). For quantitative evaluation, we considered the manual annotations as ground truth, and assessed the accuracy using the adjusted Rand index (ARI). Higher ARI scores were achieved by STitch3D's multislice result compared to its single-slice result, indicating STitch3D's ability to borrow information across slices (Fig. 2e). We compared STitch3D with spatial domain identification methods including BayesSpace[12], SpaGCN[13] and STAGATE[14]. Unlike STitch3D, these methods showed less satisfactory capability in the detection of cortical layers (Fig. 2e and Supplementary Fig. 1). The advantage of STitch3D's 3D spatial region identification is due to its effective integration of slices in the shared latent space (Fig. 2d). The latent representations showed consistent patterns across all slices, with a trajectory from WM and L6 to L1 (refs. 20,21) (Supplementary Fig. 2). The recovered spatial trajectory corresponds to corticogenesis, during which cortical neurons are born successively from inner to outer layers[22].

Next, we evaluated STitch3D's cell-type deconvolution performance. We used a corresponding cell-type reference[23], where ten excitatory neuronal subtypes were annotated. The layer specificity of each subtype was determined based on the layer marker genes. We assessed the reliability of cell-type deconvolution based on the receiver operating characteristic (ROC) analysis, in which the estimated proportions of neuronal subtypes were used to recover the layer specificity. A higher value of the area under the curve (AUC) of ROC indicates a more reliable performance. STitch3D achieved higher AUC scores in the multislice analysis compared to single-slice results (Fig. 2f). Taking the neuronal subtype named Ex_8_L5_6, which has high expressions of layers 5–6 marker genes, as an example, STitch3D recovered its clear enrichment in layers 5–6 (Supplementary Figs. 3 and 4). Consistent with the quantitative evaluation, decreased and less noisy Ex_8_L5_6 proportions in layers 1–4 were observed in STitch3D's multislice result, compared to single-slice analysis (Supplementary Fig. 4). STitch3D's multislice result enables clear characterization of the neuronal subtypes' 3D spatial distributions (Fig. 2c). We also compared STitch3D with cell-type deconvolution methods such as RCTD[15], Cell2location[16], Tangram[17], DestVI[18] and CARD[19]. These methods showed lower AUC scores compared to STitch3D (Fig. 2f). Among them, Cell2location also allows for multislice analysis. Nevertheless, it showed limited ability to borrow information across slices, as indicated by the similar AUC scores in single-slice and multislice experiments.

We also conducted a comprehensive comparison among cell-type deconvolution methods, following a benchmarking study[24]. We simulated ST slices in three different scenarios. As the ground-truth cell-type proportions in simulated datasets are known, we quantitatively measured the results using an overall accuracy score. Details are provided in the Methods and Supplementary Section 1.

In the first scenario, a mouse cortex seqFISH+ dataset[25] was used to create a simulated ST slice by gridding cells into spots (Supplementary Fig. 5a). With a corresponding single-cell reference[26], STitch3D achieved the highest overall accuracy score among all methods (Fig. 2g and Supplementary Fig. 5b). Using this dataset, we also validated STitch3D's gene imputation accuracy (Supplementary Fig. 5c). Compared to Tangram, which is also capable of gene imputation, STitch3D achieved better scores for all three testing genes (Supplementary Fig. 5d). In the second scenario, we simulated multiple slices from a mouse visual cortex STARmap slice[27] as technical replicates by adding independent noises (Supplementary Fig. 6a). In single-slice experiments, STitch3D was among the best two methods (Fig. 2h). When more slices were incorporated, STitch3D showed stably improved accuracy, with its result becoming less noisy when including more slices (Supplementary Fig. 6b). In the third scenario, we used multiple MERFISH slices of the mouse hypothalamic preoptic region[28] (Supplementary Fig. 7). Among the compared methods, STitch3D showed the best performance for a single slice (Bregma −0.24; Fig. 2i). When jointly analysing three adjacent slices (Bregma −0.29, −0.24, −0.19), STitch3D achieved an improved overall score. Similar analyses were performed using the other two sets of three adjacent slices, which showed consistent results (Supplementary Fig. 8).

https://doi.org/10.1038/s42256-023-00734-1

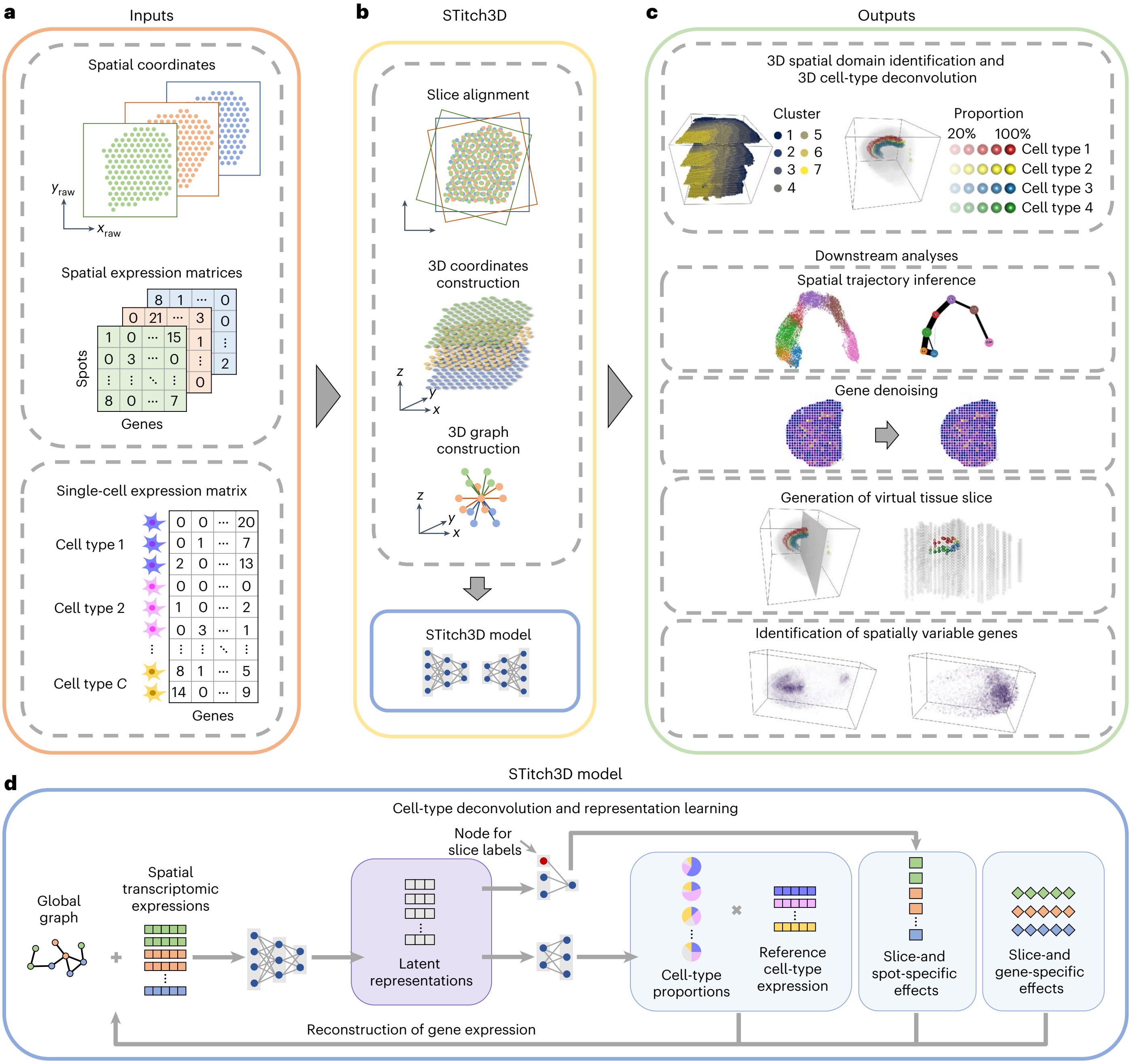

Fig. 1 | Overview of STitch3D. a, Raw data from multiple ST tissue slices, and cell-type-specific gene-expression profiles from a reference scRNA-seq dataset, are taken as STitch3D's inputs. **b**, The preprocessing steps of STitch3D include alignment of spots from different tissue slices to build 3D locations of spots, and construction of a global 3D graph. The main model of STitch3D incorporates these structures to perform representation learning for 3D spatial domain identification and 3D cell-type deconvolution. **c**, STitch3D outputs the 3D spatial region identification result and the estimation of 3D spatial distributions of different cell types in tissues. STitch3D also enables multiple downstream analyses, including spatial trajectory inference, denoising of low-quality gene-expression measurements, generation of virtual tissue slices and identification of genes with 3D spatial expression patterns. **d**, STitch3D jointly models multiple slices, and utilizes a graph attention-based neural network to learn latent representations of spots and cell-type proportions with 3D spatial information.

Besides the above experiments, STitch3D also showed good performance when applied to high-resolution datasets, suggesting its high flexibility and wide applicability (Supplementary Section 7 and Supplementary Figs. 9–11).

In summary, using the above experiments, we have shown STitch3D's state-of-the-art performance in single-slice analysis. More importantly, STitch3D's multislice analysis further enables it to achieve consistent results with higher accuracy, serving as a foundation for 3D reconstruction.

Reconstruction of the adult mouse brain

In this section we demonstrate that STitch3D can accurately reconstruct a complex 3D adult mouse brain. We used 35 coronal slices across the antero–posterior (AP) axis[5] and a cell-type reference containing 59 cell types[16]. The 3D reconstruction task here is challenging, because it requires methods to account for batch effects in dozens of slices, and to distinguish subtle difference among nuanced cell subtypes.

We applied STitch3D after verifying the correctness of slice alignment (Supplementary Section 3 and Supplementary Figs. 12–14).

 https://doi.org/10.1038/s42256-023-00734-1

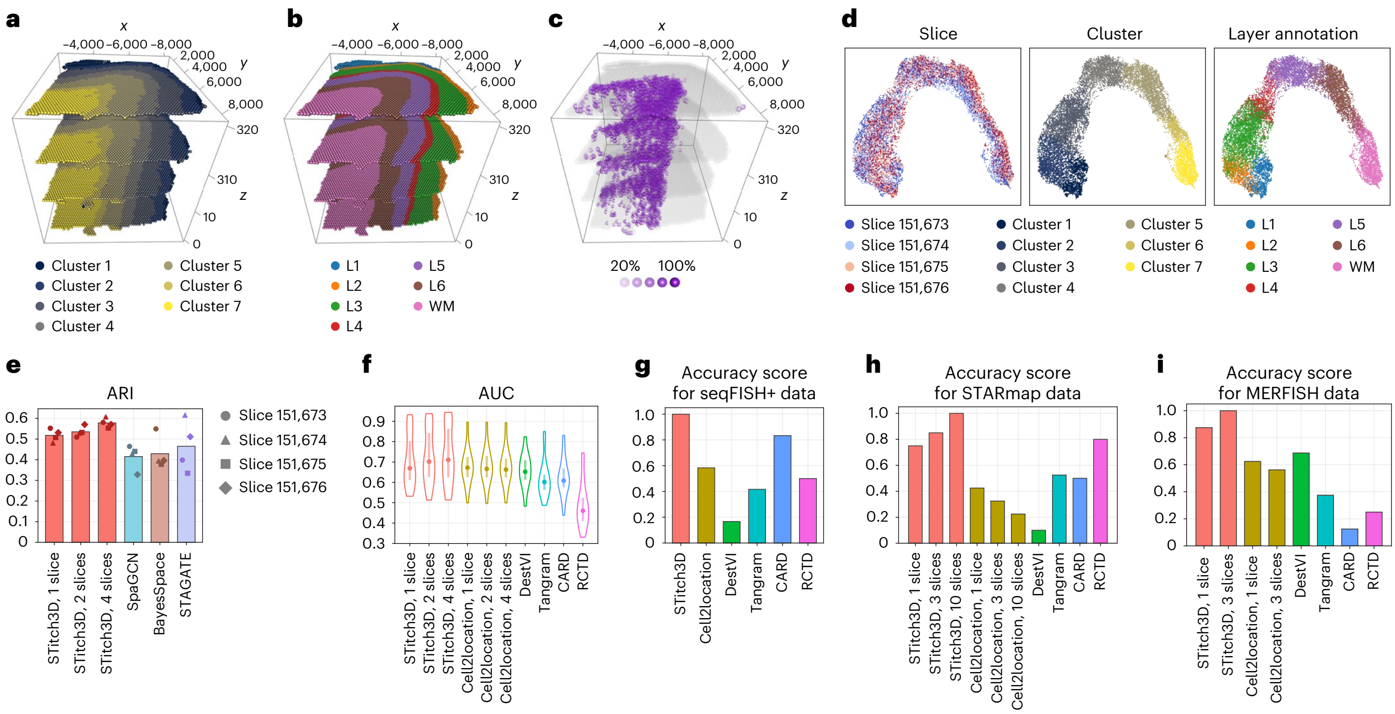

Fig. 2 | Benchmarking of STitch3D and other state-of-the-art methods. **a**, STitch3D's 3D spatial domain detection result for the DLPFC data. **b**, 3D visualization of manual annotations. **c**, STitch3D's reconstructed 3D spatial distribution of the Ex_8_L5_6 neuronal subtype. Spots with proportion values larger than 20% are shown in purple, and a lower transparency indicates a higher proportion. **d**, We visualized STitch3D's learned representations of spots in the shared latent space with UMAP plots[63], coloured by slice indices, STitch3D's assigned domain labels and manual annotations. **e**, ARI scores of the spatial domain detection methods. **f**, Violin plots of the AUC scores of cell-type deconvolution methods measured with ten region-specific neuronal subtypes. Dots indicate median values, and bars indicate the range between the 25th and 75th percentiles, calculated based on $n = 40$ points. **g**, Overall accuracy scores of the compared methods in a seqFISH+ data-based simulation study. **h**, Overall accuracy scores of the compared methods in a STARmap data-based simulation study. **i**, Overall accuracy scores of the compared methods in a MERFISH data-based simulation study.

STitch3D partitioned the brain into well-organized 3D domains based on integrated spot representations (Supplementary Fig. 15a). For example, three layer-structured domains labelled as clusters 1, 2 and 5 formed the isocortex region (Fig. 3a–c and Supplementary Fig. 16). These clusters showed strong connectivity in trajectory inference (Supplementary Fig. 15b). Pseudotime analysis within these clusters then showed a consistent pattern with corticogenesis across all slices (Fig. 3j). Clusters 3 and 9 correspond to the hippocampus and thalamus regions, which change along the AP axis, indicating STitch3D's ability to preserve biological variation among slices (Supplementary Figs. 17d,e and 18). Using this example, we also verified the effectiveness of STitch3D's batch-effect modelling. When we removed it from STitch3D, the slices were less well-integrated, leading to inconsistent spatial domain detection across slices (Supplementary Fig. 17c–e).

Leveraging fine-grained cell-type signatures from the reference, STitch3D revealed 3D cell-type distributions. For example, it accurately reconstructed distributions of four hippocampal neuron types in cornu ammonis (CA) and dentate gyrus (DG) (Fig. 3e,f). These distributions correctly matched hippocampus areas CA1, CA2, CA3 and DG annotated in the Allen Reference Atlas – Mouse Brain[29] (Fig. 3d and Supplementary Fig. 19). STitch3D also captured distributions of excitatory neurons across cortex layers (Supplementary Fig. 20), and other major regional subtypes (Supplementary Fig. 21). For quantitative comparisons among methods, we used estimated proportions of four hippocampal neuron types to recover the CA1, CA2, CA3 and DG regions and compared the result with region annotations using ROC analysis. A similar analysis was conducted with the two cortical neuron types in layers 2–3 and layers5–6 (Fig. 3g). STitch3D's single-slice analysis already showed overall better AUC scores compared to the other methods. By borrowing information across slices, STitch3D achieved better accuracy in its multi-slice analysis (Supplementary Fig. 22). We visualized the estimated 3D cell-type distributions in single-slice and multislice analyses. Cell-type distributions of layers 5–6 and DG1 excitatory neuronal subtypes estimated by STitch3D's single-slice analysis showed noisy scattering patterns outside the annotated areas (Supplementary Fig. 23c,d). In comparison, these cell types clearly concentrated in the expected areas in the multislice analysis (Supplementary Fig. 23a,b). We also varied the number of input slices from 20 to 35 to investigate its impact, and observed STitch3D's satisfactory 3D reconstruction with different slice numbers. Besides, it showed robust performance with varying data quality. More details are included in Supplementary Sections 4 and 6 and Supplementary Figs. 24–28.

One problem in ST studies is the extensive noise in gene-expression measurements[30]. STitch3D enables gene denoising by combining estimated cell-type proportions with cell-type signatures from the reference. We demonstrated this function with the genes *Tle4* and *Rell1*, which have noisy patterns in raw data. After applying STitch3D, the gene-expression patterns became clear and consistent with the Allen Mouse Brain Atlas[31] (Fig. 3h,i and Supplementary Fig. 29), validating STitch3D's reliability.

STitch3D's 3D reconstruction results further provide auxiliary views of tissues that are not captured by original slices. With the reconstructed 3D model using coronal slices, we created a virtual sagittal slice by introducing a plane parallel to the AP axis and projecting adjacent spots onto it (Fig. 3k). With identified spatial domains and estimated cell-type proportions, the virtual slice correctly revealed the sagittal brain structure (Fig. 3l), showing a consistent pattern with the sagittal slice reference in the Allen Reference Atlas[29] (Supplementary Fig. 30).

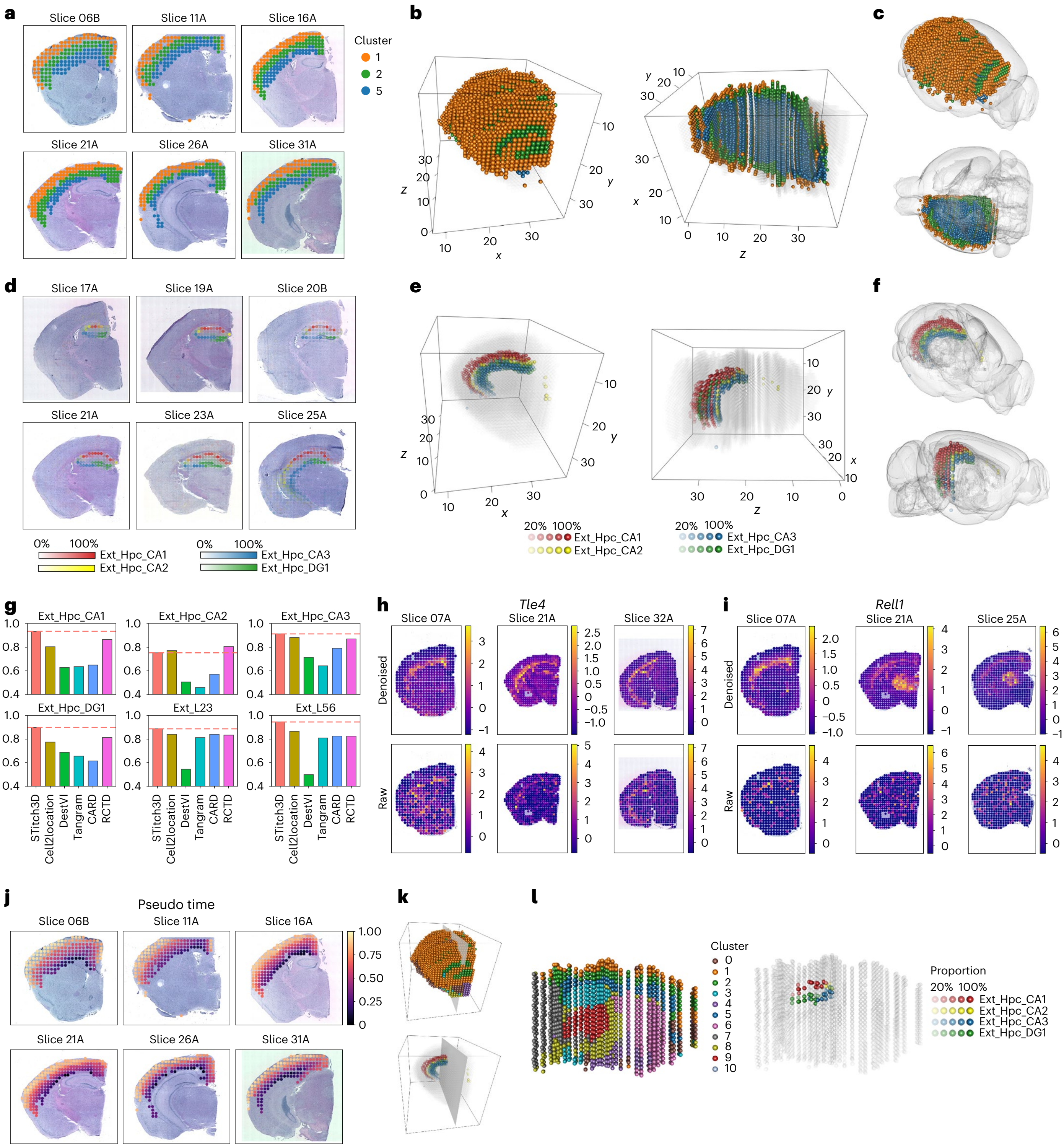

Fig. 3 | 3D reconstruction of the adult mouse brain. **a**, Clusters 1, 2 and 5 in STitch3D's spatial domain detection result visualized on 2D ST slices. **b**,**c**, 3D visualizations of clusters 1, 2 and 5 in STitch3D's aligned 3D coordinates using ICP (**b**) and in the Allen Mouse Brain Common Coordinate Framework (CCFv3)[64] (**c**). **d**, Estimated proportions of four hippocampal neuron types in STitch3D's cell-type deconvolution result visualized on 2D ST slices. A lower transparency indicates a higher proportion. **e**,**f**, 3D visualizations of the proportions of four hippocampal neuron types in STitch3D's aligned 3D coordinates using ICP (**e**) and in CCFv3 (**f**). Spots with proportion values larger than 20% are shown. **g**, AUC scores of the compared methods measured with six regional cell types. Horizontal dashed lines indicate STitch3D's scores. **h**,**i**, Denoised and raw expression levels of genes *Tle4* (**h**) and *Rell1* (**i**) in *z*-scores. **j**, Spatial trajectory among cortex layers (clusters 1, 2, 5) with pseudotime visualized on six 2D slices. **k**, Visualization of the plane containing the virtual sagittal slice. **l**, The virtual sagittal slice with spatial regions, and the proportions of four hippocampal neuron types, respectively.

Reconstruction of the developing human heart

We applied STitch3D to a human heart ST dataset, including three sets of slices collected at 4.5–5, 6.5 and 9 post-conception weeks (PCWs)[7]. We first focused on the 6.5-PCW human heart, in which five clusters were consistently identified across nine slices by STitch3D (Fig. 4a,b). The outflow tract (OFT) and the atria were detected by clusters 3 and 4, respectively. Clusters 1, 2 and 5 identified different ventricle regions. For the three ventricle clusters, we performed gene ontology (GO) analysis[32] to characterize their functionality (Fig. 4d). Cluster 1 corresponds to the trabecular ventricular myocardium region. Its top enriched GO terms mostly relate to cellular respiration. Cluster 2 corresponds to compact ventricular myocardium and epicardium, and is enriched with processes related to heart development and heart contraction. Cluster 5 corresponds to areas containing cavities in the ventricles, and is enriched with gas transport terms and cellular response to metal ions. The inferred trajectory from clusters 5, 1 to cluster 2 showed the developmental process of the early trabecular ventricular myocardium to form the ventricular myocardial wall[33] (Supplementary Fig. 31).

STitch3D also accurately generated 3D cell-type distributions. Among three cardiomyocyte subtypes, atrial cardiomyocytes (ACMs) and ventricular cardiomyocytes (VCMs) were correctly found localized in the atria and ventricles (Fig. 4c,f,k). *Myoz2*-enriched cardiomyocytes were found in both the atria and ventricles (Fig. 4c,k), which is consistent with the original study[7]. Furthermore, enrichment of smooth muscle cells (SMCs) was shown in the OFT (Fig. 4c,g,k), validated by gene marker *MYH11* (Supplementary Fig. 32). As supported by gene marker *TBX18*, the 6.5-PCW heart was correctly found surrounded by epicardium-derived cells (EPDCs; Fig. 4g and Supplementary Fig. 32). For cell-type colocalizations, ACMs and VCMs distribute in separate areas, but are all colocalized with *Myoz2*-enriched cardiomyocytes (Fig. 4e). Although both being endothelial cell types, capillary endothelium is colocalized with VCMs, whereas endothelium/pericytes is mainly colocalized with EPDCs. With model extensions, STitch3D also enables mapping single cells to spatial locations in the heart (Supplementary Section 8 and Supplementary Fig. 33).

For quantitative evaluation, according to the single-cell analysis in the original study, ACMs, cardiac neural crest cells and SMCs were found absent in the ventricles, serving as a guideline to evaluate the result reliablity. Compared to other methods, which sometimes overestimated these three cell-type proportions in the ventricles, STitch3D produced reasonable cell-type deconvolution results (Fig. 4i,k and Supplementary Fig. 34). Moreover, the original study identified marker genes *MYH6*, *MYH7* and *MYOZ2* for ACMs, VCMs and *Myoz2*-enriched cardiomyocytes, respectively. DestVI's estimated cell-type proportions of ACMs and VCMs are less consistent with marker gene expressions, as indicated by lower Pearson correlation scores (Fig. 4j). CARD showed less satisfactory *Myoz2*-enriched cardiomyocyte proportions. Compared to the other methods, STitch3D produced more accurate estimations.

After verifying the reliability of STitch3D's reconstructed 6.5-PCW human heart, we applied STitch3D to 4.5–5-PCW and 9-PCW hearts. We observed similar distribution patterns of ACMs, VCM, SMCs and fibroblast-like cells in all stages (Fig. 4k and Supplementary Fig. 35). For detailed temporal differences, EPDCs were found to be enriched on the 6.5- and 9-PCW heart surfaces, but they showed a relatively lower density on the 4.5–5-PCW heart surface (Fig. 4g,h and Supplementary Figs. 32, 36 and 37). Compared to late stages, the 4.5–5-PCW heart presented a higher density of epicardial cells in its surrounding layer. Next, we investigated differences between the 6.5- and 9-PCW hearts. We performed a *t*-test on the gene expressions in EPDC-enriched spots detected by STitch3D, and found that EPDC-enriched spots in the 6.5-PCW heart showed higher expression levels of gene *IGF2* (Fig. 4l). In a similar analysis of SMC-enriched spots, genes *ACTB* and *FOS* showed higher expression levels in the 6.5-PCW and 9-PCW heart, respectively (Fig. 4m,n). These nuanced temporal differences identified using STitch3D help to better characterize the human cardiogenesis process.

Analysis of HER2-positive breast cancer data

Besides normal tissue data, we applied STitch3D to murine lymph node data after *Mycobacterium smegmatis* stimulation[18] and mouse skin data during wound healing[34,35], to demonstrate its ability to capture the cellular-level response to perturbation (Supplementary Section 5 and Supplementary Figs. 38 and 39). In this section, we analysed HER2-positive breast tumour data[36,37] to show STitch3D's ability in relation to biological findings in abnormal tissues.

After applying STitch3D to tumour samples from eight patients, global cell-type colocalizations were calculated. We found some consistent results in all samples, for example, the negative spatial correlation between two cancer-associated fibroblasts (CAFs; Supplementary Fig. 40). Additionally, mesenchymal stem cells and inflammatory-like CAFs (MSC/iCAF-like) were found colocalized with immune cells including B cells, T cells and plasma cells, showing a different pattern compared to myofibroblast-like CAFs (myCAF-like).

We then focused on results for patient E. Among the three slices, the first slice was annotated by pathologists. Using STitch3D, the three slices were well-integrated (Fig. 5a), which allowed us to transfer pathologists' annotations to unlabelled slices (Fig. 5b). STitch3D also identified six spatial areas (Fig. 5a,d). Specifically, clusters 2–5 recovered invasive cancer regions, and STitch3D indeed found cancer epithelial cells enriched in these clusters compared to the other areas (Fig. 5e,f). The high correlation between the cancer epithelial cell proportion and breast cancer marker *ERBB2* expression supported STitch3D's result (Fig. 5c). Furthermore, the distributions of the two CAF subtypes showed different distribution patterns (Fig. 5e), which is consistent with their negative cell-type proportion correlation. Next, we investigated the heterogeneity in cancer regions. Both clusters 4 and 5 showed enrichment of cancer epithelial cells. However, cluster 4 showed higher enrichment of B cells and T cells, while cluster 5 showed higher enrichment of cancer epithelial cells (Fig. 5f). We performed differential expression analysis followed by GO analysis for these clusters (Fig. 5g,h). Cluster 4 showed enrichment of the immune system process, which validated STitch3D's identified enrichment of immune cells. In cluster 5, we observed enrichment of the response to endoplasmic reticulum stress, which plays an important role in breast cancer progression[38]. Cluster 5 also showed enrichment of the regulation of apoptotic process. The distinctions in enriched biological processes indicated differences between these two regions, highlighting their respectively contrasting immune/stromal microenvironments.

Establishment of a *Drosophila* embryo whole organism atlas

We applied STitch3D to reconstruct a 3D *Drosophila* embryo model to demonstrate its capability to reconstruct a 3D whole organism spatial atlas. We analysed the 16–18 h embryo profiled by Stereo-seq[8], in which spatial bins were constructed by merging 20 × 20 DNA nanoballs in the original study. To apply STitch3D, we built a cell-type reference based on a scRNA-seq *Drosophila* embryo atlas[39] at the same developmental stage. The cell-type annotation is based on marker genes provided by the Berkeley *Drosophila* Genome Project (BDGP)[40,41] and FlyBase[42] (Supplementary Table 1 and Supplementary Figs. 41 and 42).

This task is challenging due to the high sparsity of *Drosophila* embryo ST data (Supplementary Fig. 43). Despite the difficulty, STitch3D successfully reconstructed the *Drosophila* embryo by integrating the ST and scRNA-seq datasets (Fig. 6 and Supplementary Figs. 44–46). For example, in STitch3D's reconstruction, central nervous system (CNS) spots assembled into a spatial structure that recapitulates embryonic CNS morphology and position[43] (Fig. 6a). The result was also validated by CNS-specific marker gene *Obp44a* (Fig. 6b). Another example is the reconstructed salivary gland, which again fully recapitulated the morphology and position of the salivary gland (Fig. 6d), and showed a similar pattern to the salivary gland-specific marker gene *CG14453* (Fig. 6e). In comparison with

https://doi.org/10.1038/s42256-023-00734-1

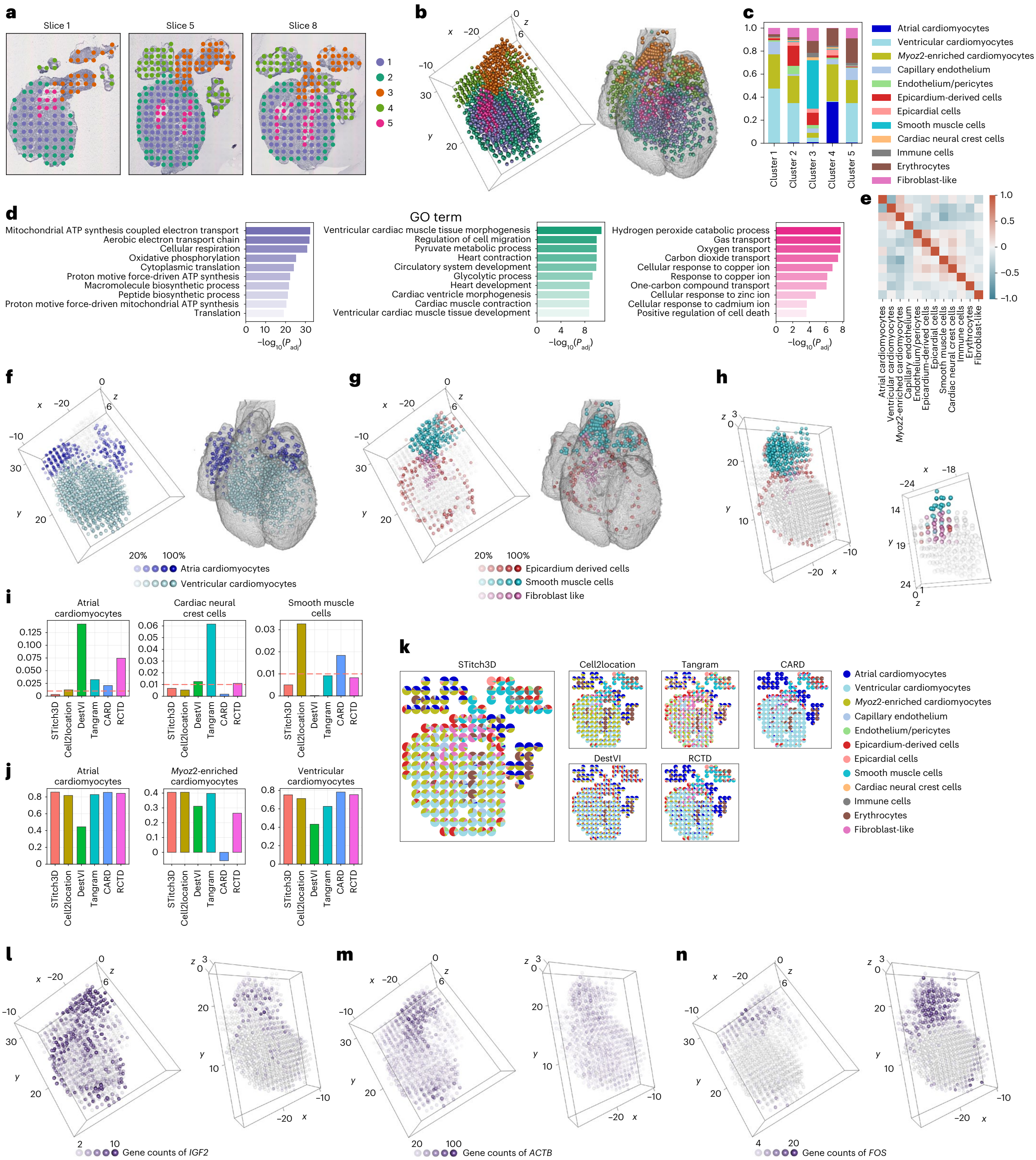

Fig. 4 | 3D reconstruction of the developing human heart. **a**, STitch3D's spatial domain detection result visualized on 2D ST slices from the 6.5-PCW heart. **b**, The detected domains visualized in STitch3D's aligned 3D coordinates of the 6.5-PCW slices, and in a heart model of Carnegie-stage-18 embryo (CS18-6524)[65]. **c**, The average proportions of cell types in the spatial domains estimated by STitch3D. **d**, GO analysis for spatial domains in the 6.5-PCW heart. **e**, Spatial colocalization of cell types in the 3D atlas of the 6.5-PCW heart. **f**, STitch3D's estimated 3D spatial distributions of ACM and VCM in the 6.5-PCW heart. Spots with proportion values larger than 20% are shown. **g**,**h**, 3D spatial distributions of EPDCs, SMCs and fibroblast-like cells in the 6.5-PCW heart (**g**), the 9-PCW heart (**h**, left) and the 4.5–5-PCW heart (**h**, right). Spots with proportion values larger than 20% are shown. **i**, Average proportions of ACM, cardiac neural crest cells and SMCs in the ventricles, estimated by the compared methods. Horizontal dashed lines indicate 0.01. **j**, Pearson correlations between estimated cell-type proportions and marker gene-expression patterns. **k**, Estimated cell-type proportions in a 6.5-PCW heart slice visualized by pie charts. **l**–**n**, Spatial expression patterns of genes *IGF2* (**l**), *ACTB* (**m**) and *FOS* (**n**).

https://doi.org/10.1038/s42256-023-00734-1

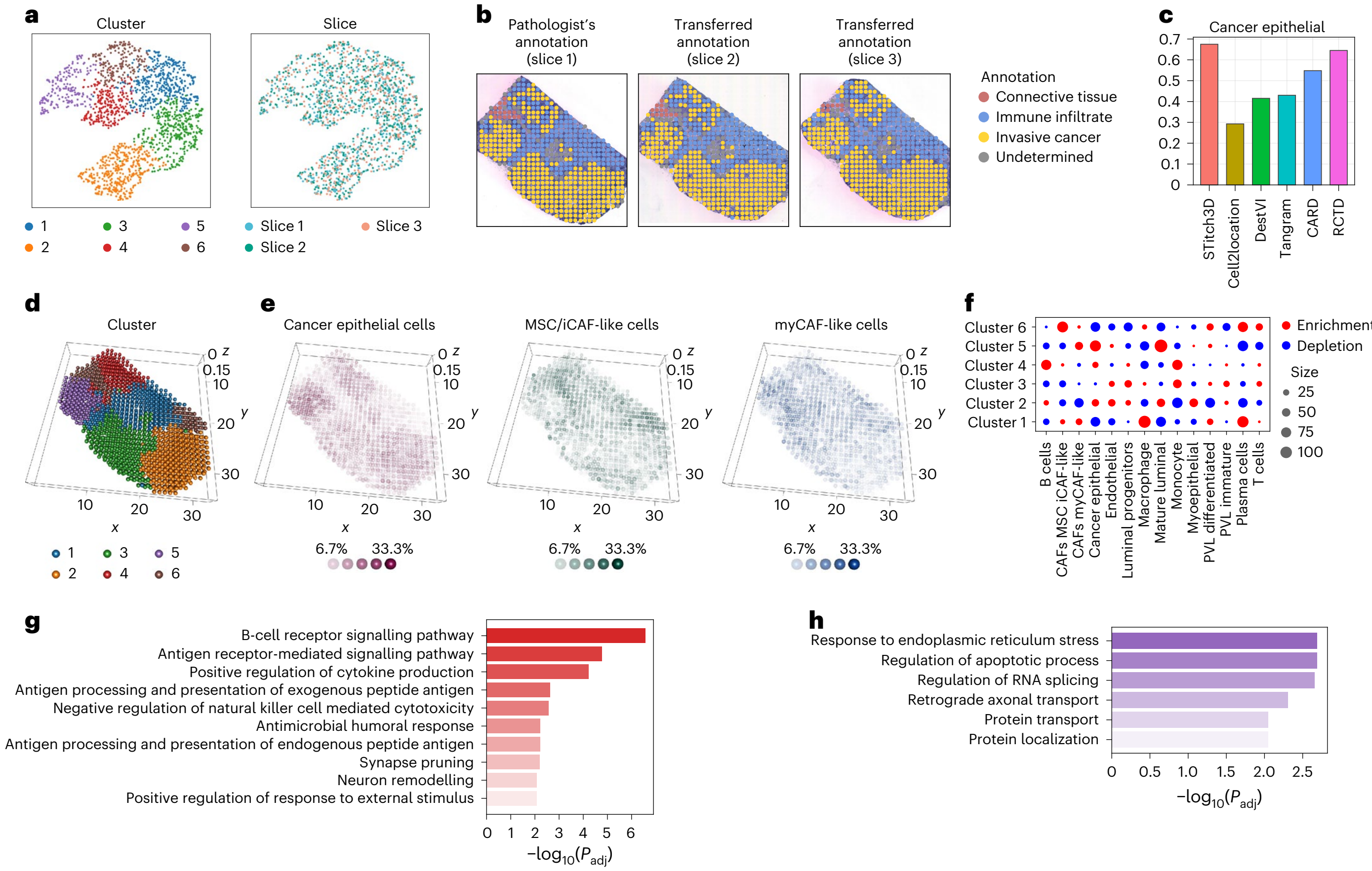

Fig. 5 | Application of STitch3D to the breast cancer data. a, UMAP plots of latent spot representations learned by STitch3D, coloured by cluster labels and slice indices. **b**, Pathologist's annotation for slice 1, and transferred annotation for slices 2 and 3 using STitch3D's learned representations. **c**, Pearson correlations between the estimated proportion of cancer epithelial cells and pattern of breast cancer marker gene *ERBB2*. **d**, STitch3D's 3D spatial domain detection result. **e**, STitch3D's reconstructed 3D spatial distribution of cancer epithelial cells and two CAF subtypes. **f**, Enrichment or depletion of different cell types in all clusters. **g**,**h**, Notable GO terms for clusters 4 (**g**) and 5 (**h**).

other cell-type deconvolution methods, STitch3D achieved the highest correlation between regional cell-type distributions and marker genes (Fig. 6c,f). We also compared STitch3D's reconstructed CNS and salivary gland regions to those annotated by the original study, and found that STitch3D's result was more similar to late-stage embryonic CNS and salivary gland morphology[44–47] (Supplementary Fig. 47). The reconstructed embryo allows us to zoom in to specific regions within an organ type and uncover novel gene-expression patterns. For example, we identified gene *CG14265* as a highly expressed gene in the salivary gland. It exhibited a medial-to-lateral expression gradient (Fig. 6g,h) that was not captured by the original slices.

Additionally, in STitch3D's result, foregut, proventriculus, midgut and hindgut/anal pad were detected in coherent spatial regions from anterior to posterior (Fig. 6i,j and Supplementary Figs. 48 and 49). However, Cell2location, CARD, DestVI and RCTD failed to detect the proventriculus or hindgut/anal pad. Although Tangram detected these regions, the spots failed to spatially assemble into a coherent structure (Fig. 6i). STitch3D's performance is also robust to varying slice numbers and resolutions (Supplementary Figs. 50–52).

Among STitch3D's identified regions, proventriculus and hindgut/anal pad were not annotated in the original study. We then performed differentially expressed gene analysis in these regions. We first focused on the proventriculus region, and used a *t*-test to find proventriculus-enriched genes[48]. We found 122 genes having adjusted *P* values less than 0.01, $\log_2$(fold change) values larger than 0.75, and non-zero counts in more than 1% of spots. In particular, the majority of these genes were verified to be localized in late-stage embryonic proventriculus or nearby regions by BDGP and Fly-FISH[49,50] (Fig. 6k). Some of the genes are not available in the databases or not found in staining. However, they are highly colocalized with known proventriculus-enriched genes (Fig. 6m). Similarly, for the hindgut/anal pad region, we obtained 72 genes with statistical significance (Fig. 6l). These genes show high concentration in the hindgut/anal pad, as identified by STitch3D (Supplementary Fig. 53). Moreover, we identified genes enriched in different subregions within the foregut and midgut (Supplementary Figs. 54–56), facilitating biologists to build a comprehensive gene regulatory network that controls the gut differentiation process. To summarize, the above results highlight the power of STitch3D in reconstructing a virtual embryo with accurate 3D regional cell-type distribution patterns, which will provide a powerful tool to study embryogenesis processes at a systemic level.

Discussion

We have presented STitch3D for characterizing 3D tissue architectures by integrating multiple ST slices to identify 3D tissue regions and 3D cell-type distributions. We have demonstrating STitch 3D's reliability using diverse datasets, with scales ranging from tissues to whole organisms.

STitch3D improves over existing methods in two main aspects. First, most tools were developed for either spatial domain detection or cell-type deconvolution tasks. However, the two tasks are inherently

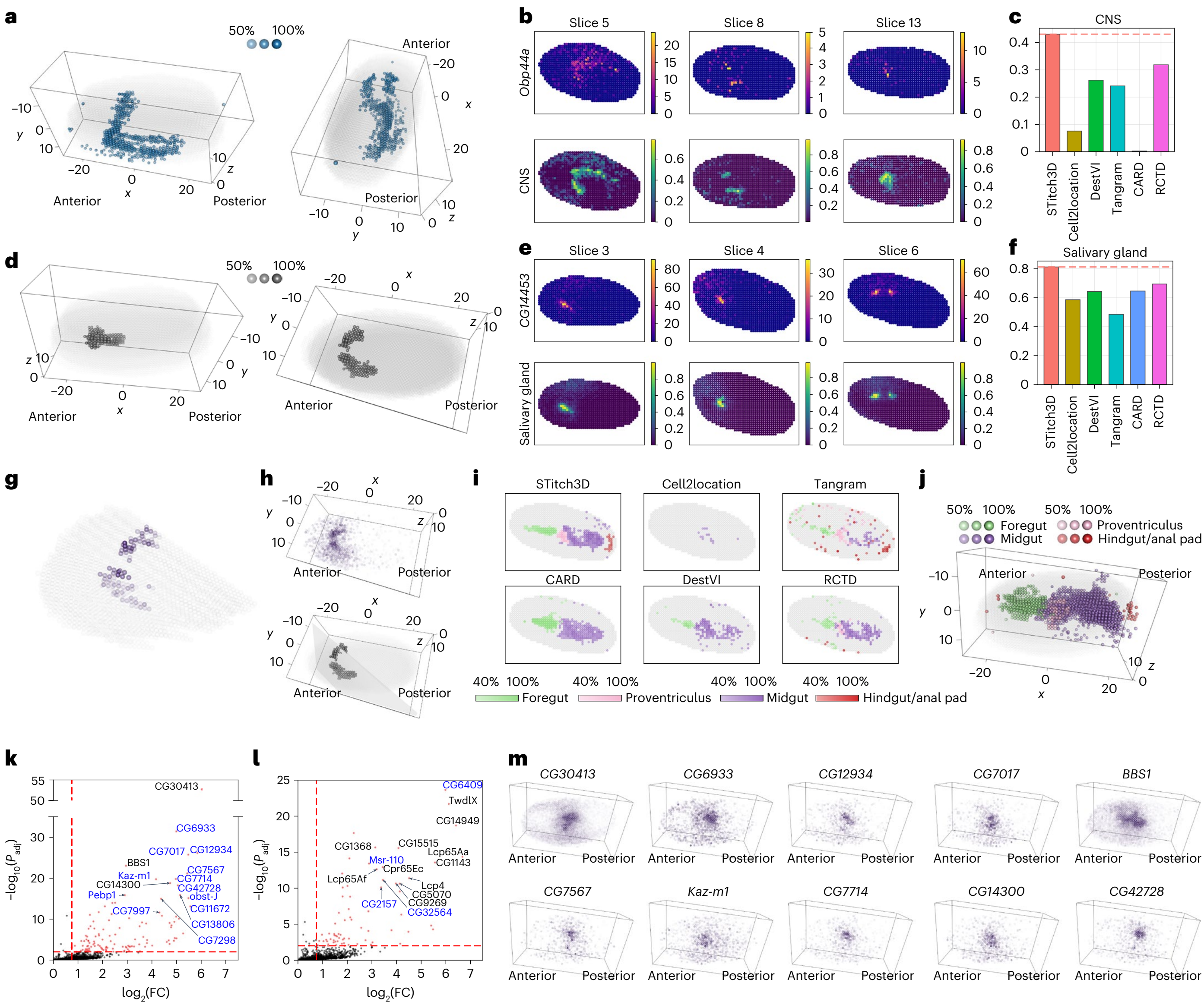

Fig. 6 | STitch3D constructed the 3D atlas of the *Drosophila* embryo. **a**,**d**, Visualizations of STitch3D's estimated 3D distributions of the CNS (**a**) and salivary gland (**d**), where spots with proportion values larger than 50% are shown. **b**,**e**, Visualizations of expression patterns of CNS-related marker gene *Obp44a* (**b**) and salivary gland-related marker gene *CG14453* (**e**). We compared the expression patterns of the marker genes with STitch3D's estimated proportions of the CNS (**b**) and salivary gland (**e**). **c**,**f**, Pearson correlations between the estimated proportions of cell types (CNS (**c**) and salivary gland (**f**)) and marker gene expressions (*Obp44a* and *CG14453*, respectively). Horizontal dashed lines indicate STitch3D's results. **g**, Visualization of the expression levels of *CG14265* in salivary gland-enriched spots on the virtual slice generated by STitch3D. **h**, 3D visualizations of the spatial expression pattern of *CG14265* (top), and salivary gland-enriched spots and the plane containing the virtual slice (bottom). **i**, We examined the capability of the compared methods to identify the foregut, proventriculus, midgut and hindgut/anal pad. Spots with proportion values larger than 40% are shown. **j**, Visualization of STitch3D's estimated 3D distributions of the foregut, proventriculus, midgut and hindgut/anal pad. Spots with proportion values larger than 50% are shown. **k**,**l**, Differentially expressed gene analysis for STitch3D's identified proventriculus region (**k**) and hindgut/anal pad region (**l**). Horizontal lines indicate a value of 2.0 (i.e. $P_{adj} = 0.01$), and vertical dashed lines indicate a value of 0.75. Known genes localized in these regions according to BDGP and Fly-FISH are annotated in blue. FC, fold change. **m**, Expression patterns of highly variable genes in STitch3D's identified proventriculus region.

connected, as cell-type compositions are often distinct across regions. STitch3D jointly handles the two tasks. Hence, the cell-type compositions can serve as augmented information for identifying biologically interpretable 3D regions. Second, the most important feature of STitch3D is its ability to integrate multiple slices for 3D reconstruction, whereas most methods were designed only for analysing one single slice. STitch3D assembles information across 3D neighbourhood spots, and provides effective batch-effect modelling. It has been shown by previous studies that data integration improves statistical power in single-cell analysis[51–54]. With model innovations, STitch3D achieved improved performance in both spatial domain detection and cell-type deconvolution by the integration of multiple slices.

In this study we have focused on using gene-expression and spatial location information. Utilization of haematoxylin and eosin (H&E)-stained histological images and multi-omics data[55] could improve the interpretability of results, and is left to future work.

3D tissue reconstruction is evidently very meaningful for accelerating biological discoveries. As more ST datasets containing multiple slices are generated, the demand for assembling 3D panoramic views of tissues will grow rapidly. STitch3D caters to this demand for tools to reconstruct 3D tissue structures, and will play an essential role in ST data analysis.

Methods

The model of STitch3D

STitch3D first incorporates the iterative closest point (ICP) algorithm[56] or PASTE[57] algorithms to align multiple slices by the registration of spatial spots. Details of the alignment step are presented in Supplementary Section 2. After the alignment of multiple slices, STitch3D assembles the 2D locations of spots along a newly introduced z axis, which depicts the distances between pairs of slices, to build 3D spatial locations of spots. With the 3D spatial locations of spots, STitch3D constructs a global 3D neighbourhood graph for spots in all slices. By default, two spots are connected with an edge in the global 3D graph if their distance is less than 1.1 times the distance between two nearest spots in one slice.

Next, we use the STitch3D model to perform 3D spatial region detection and 3D cell-type deconvolution. STitch3D performs 3D analyses by integrating information from multiple slices of an ST dataset and an scRNA-seq reference dataset. In the ST data, we use $s = 1, 2, ..., S$ as the index of tissue slices. For slice s, we denote its number of spots as N_s, and the observed gene-expression count matrix as $Y^s = [Y^s_{n,g}] \in \mathbb{R}^{N_s \times G}$, with index $n = 1, 2, ..., N_s$ for spots and $g = 1, 2, ..., G$ for genes. In the annotated scRNA-seq references, we denote the obtained cell type-specific gene-expression profile as matrix $V = [V_{c,g}] \in \mathbb{R}^{C \times G}$, where $c = 1, 2, ..., C$ is the index of cell types. In this matrix, the row vector $\mathbf{v}_{c,\cdot} \in \mathbb{R}^G$, which satisfies $\sum_g V_{c,g} = 1$, characterizes the mean expression signature of cell type c.

To integrate both gene-expression levels and spatial locations across all tissue slices for 3D analyses, STitch3D first encodes the gene-expression information of spots into a shared latent space, with a neural network incorporating our constructed 3D spatial locations. For clarity, we concatenate the index for spots across slices $s = 1, 2, ..., S$ to introduce a new global index $i = 1, 2, ..., N$, for spots in all slices where $N = N_1 + N_2 + \cdots + N_S$. We denote $A = [A_{i,j}] \in \mathbb{R}^{N \times N}$ as the global adjacency matrix specifying the 3D global graph, where $A_{i,j} = 1$ if there exists an edge between spot i and spot j, that is, spot i and spot j are 3D neighbouring spots, and $A_{i,j} = 0$ otherwise. Let $Y = [Y_{i,g}] \in \mathbb{R}^{N \times G}$ be the concatenated gene-expression count matrix across all slices. Specifically, the latent representations of spots are obtained by

$$\begin{aligned} X_{i,g} &= \log\left(\frac{Y_{i,g}}{\sum_{g=1}^{G} Y_{i,g}} \times 10^4 + 1\right), \\ Z &= f_Z(X, A) \end{aligned}$$

where we perform normalization and log-transformation on count matrix Y for network stability, $X = [X_{i,g}] \in \mathbb{R}^{N \times G}$ is encoded to $Z \in \mathbb{R}^{N \times p}$ through a graph attention network $f_Z(\cdot)$ and p is the dimensionality of the shared latent space. Clearly, the latent representation Z encodes information from both gene-expression X and the spatial neighbours given in graph A.

STitch3D then generates cell-type proportions based on latent representations of spots. We denote the proportion of cell type c in spot i as $\beta_{i,c}$. Let $\beta_i = [\beta_{i,1}, \beta_{i,2}, \cdots, \beta_{i,C}]^T \in \mathbb{R}^C$ with $\beta_{i,c} \geq 0$, $\sum_{c=1}^{C} \beta_{i,c} = 1$ represent cell-type proportions for spot i, and $Z_i \in \mathbb{R}^p$ be the obtained latent representation for spot i, where $i = 1, 2, \cdots, N$. We assume that cell-type proportions β_i can be related to latent representation Z_i through a neural network $f_\beta(\cdot)$ as $\beta_i = f_\beta(Z_i)$. Of note, with graph attention network $f_Z(\cdot)$, the obtained latent codes Z are aware of 3D global spatial information. Consequently, as β_i is generated from Z_i, spatial location information is also incorporated into the estimated cell-type proportions β_i.

To account for technical effects between ST data and scRNA-seq data, as well as batch effects among multiple ST slices, STitch3D further introduces two effects α_i^s and γ_g^s to its model, where $\alpha_i^s \in \mathbb{R}$ represents slice- and spot-specific effects, and $\gamma_g^s \in \mathbb{R}$ represents slice- and gene-specific effects. Combining cell-type-specific gene-expression profile matrix $V = [V_{c,g}]$, cell-type proportions $\beta_i = [\beta_{i,c}]$, as well as the two effects α_i^s and γ_g^s, STitch3D is able to reconstruct the observed counts in ST data with the following model:

$$\begin{aligned} Y_{i,g} &\sim \text{Poisson}\left(l_i \lambda_{i,g}\right), \\ \lambda_{i,g} &= \exp\left[\log\left(\sum_{c=1}^{C} \beta_{i,c} V_{c,g}\right) + \alpha_i^s + \gamma_g^s\right] \end{aligned}$$

where l_i is the observed total transcript count in spot i. We could also adopt a negative binomial (NB) distribution to model $Y_{i,g}$ as an alternative. We set STitch3D with the Poisson distribution as the default setting, and used this version throughout the manuscript. Empirically, we found that our STitch3D models based on the Poisson distribution and the NB distribution have comparable performance. More details about STitch3D with the NB distribution and its results are included in Supplementary Section 9 and Supplementary Figs. 57–60.

We generate slice- and spot-specific effects α_i^s from the shared latent space by using slice-specific neural network:

$$\alpha_i^s = f_\alpha(Z_i, s)$$

where s is the slice label. We model α_i^s in such a way, because we assume slice- and spot-specific effects α_i^s are related to both gene-expression information, which is distilled in Z_i, and slice label information, contained in s. We model slice- and gene-specific effects γ_g^s as trainable parameters. By accounting for unwanted variations in α_i^s and γ_g^s, β_i and Z_i can better capture biologically meaningful variation. Based on this model, the loss function is given by

$$L = -\frac{1}{N} \sum_{i=1}^{N} \sum_{g=1}^{G} \left[Y_{i,g} \log(l_i \lambda_{i,g}) - l_i \lambda_{i,g}\right]$$

In our STitch3D model, we consider a regularizer to encourage the preservation of biological variations across multiple slices. Specifically, we design the regularizer as

$$R_{\mathrm{AE}} = \frac{1}{N} \sum_{i=1}^{N} \| f_X(Z_i, s) - X_i \|_2$$

where slice-specific network $f_X(\cdot, s)$ is used to reconstruct X_i based on Z_i and slice label s. The two networks, $f_Z(\cdot)$ and $f_X(\cdot, s)$, together form an auto-encoder structure between X and Z. In regularizer R_{AE}, the slice-specific network $f_X(\cdot, s)$ is designed to account for batch effects among slices, therefore encouraging slice-shared network $f_Z(\cdot)$ to distil biological information in Z. The final objective function in STitch3D is given by $L + k_{\mathrm{AE}} R_{\mathrm{AE}}$, where k_{AE} is the coefficient for regularizer R_{AE} and is fixed at 0.1.

With the above designs, STitch3D applies stochastic gradient descent to update trainable parameters in its model. After training, STitch3D outputs latent representations Z_i and cell-type proportions β_i. The latent representations are utilized for the 3D spatial domain identification task with community detection algorithms, such the Louvain algorithm[58] and Gaussian mixture model-based clustering algorithm. Combining with the obtained 3D spatial locations, such identified clusters reveal 3D spatial regions across all slices. Meanwhile, cell-type proportions β_i show the 3D spatial distributions of different cell types, providing a comprehensive view for ST studies in a higher spatial resolution.

Selection of genes and construction of cell-type-specific gene-expression profile matrix

STitch3D selects informative genes based on the scRNA-seq reference data. Specifically, for each cell type in the annotated scRNA-seq data, STitch3D uses a t-test to find its top K marker genes with default setting $K = 500$. The top marker genes for all cell types are then concatenated to obtain a full list of informative genes that are used by STitch3D.

With G selected genes from C cell types, STitch3D constructs the cell-type-specific gene-expression profile matrix $V \in \mathbb{R}^{C\times G}$. First, STitch3D normalizes gene-expression measurements for each cell in the reference scRNA-seq dataset, such that the sum of gene expressions of each cell equals 1. Then, the row vector $\mathbf{V}_{c,\cdot}$, the mean expression signature of cell type c, is computed by averaging all normalized expressions of cells with cell type c. If different batches are included in the reference scRNA-seq dataset, STitch3D computes a cell-type-specific gene-expression profile matrix for each batch, and then takes the average of them to obtain the overall cell-type-specific gene-expression profile matrix V.

Network structures

For the encoder network $f_Z(\cdot)$, which extracts latent representations of spots in the shared latent space, STitch3D adopts a graph attention network to incorporate the constructed 3D spatial graph. The graph attention network contains a graph attention layer and a dense layer. Inputs to the graph attention layer in $f_Z(\cdot)$ are global adjacency matrix A, and normalized and log-transformed gene expression X. The dimensionality of X is the number of selected highly variable genes G, which varies from dataset to dataset. The dimensionality of the output to graph attention layer is pre-fixed at 512. Then the dense layer in $f_Z(\cdot)$ takes the 512-dimensional hidden vectors as input, and outputs 128-dimensional latent representations Z. For network $f_\beta(\cdot)$, which produces cell-type proportions, STitch3D adopts a one-layer dense network to map 128-dimensional latent representations Z to C-dimensional outputs, where C is the number of cell types, with the softmax activation. STitch3D adopts a one-layer dense network for $f_\alpha(\cdot, s)$, which takes latent representations Z and slice labels as inputs, and generates slice- and spot-specific effects α_n^s. For the slice-specific decoder network $f_X(\cdot, s)$ in the regularizer R_{AE}, a graph attention network with hidden dimensionality 512 is used. Specifically, inputs to the graph attention layer are global adjacency matrix A, latent representations Z and slice labels. The dimensionality of the output to the graph attention layer is 512. A dense layer is then used, which takes the 512-dimensional hidden vectors as input, and reconstructs the gene expression matrix.

To show how to leverage spatial location information in STitch3D, we describe details of graph attention layers used in the encoder network $f_Z(\cdot)$ and the decoder network $f_X(\cdot, s)$. Let embedding $\mathbf{u}_i^{\mathrm{in}}$ of spot i be the input to the graph attention layer. The graph attention layer updates spot embedding by using global adjacency matrix $A = [A_{i,j}] \in \mathbb{R}^{N\times N}$ of the constructed 3D global graph and the graph attention mechanism:

$$\begin{aligned}
\mathbf{u}_i^{\mathrm{out}} &= \sigma\left(\sum_{j=1}^{N} a_{i,j} W \mathbf{u}_j^{\mathrm{in}}\right), \\
a_{i,j} &= \frac{A_{i,j}\exp(e_{i,j})}{\sum_{h=1}^{N} A_{i,h}\exp(e_{i,h})}, \\
e_{i,j} &= \mathrm{sigmoid}\left(v_1^T W \mathbf{u}_i^{\mathrm{in}} + v_2^T W \mathbf{u}_j^{\mathrm{in}}\right)
\end{aligned}$$

where $\sigma(\cdot)$ is the activation function, W is the network parameter in the graph attention layer, and v_1 and v_2 are the parameters related to the graph attention mechanism. Parameters v_1 and v_2 are used to learn edge weights $a_{i,j}$ between 3D neighbourhood spots during training, helping to borrow information from neighbourhood spots adaptively for 3D analyses. For encoder network $f_Z(\cdot)$, $\mathbf{u}_i^{\mathrm{in}}$ is gene expression-level data X_i, and $\mathbf{u}_i^{\mathrm{out}}$ is a hidden vector with dimensionality 512. For decoder network $f_X(\cdot, s)$, $\mathbf{u}_i^{\mathrm{in}}$ is latent representation Z_i, and $\mathbf{u}_i^{\mathrm{out}}$ is a hidden vector with dimensionality 512.

Model training details

STitch3D utilizes Adamax, which is a variant of the Adam algorithm[59], for stochastic optimization in its model training. By default, the number of optimization steps in STitch3D is set to 20,000 with learning rate lr = 0.002, coefficients for computing running averages $\beta_1 = 0.9$, $\beta_2 = 0.999$ and weight decay parameter $\lambda = 0$. As demonstrated in Supplementary Fig. 61, STitch3D's training process guarantees the convergence of objective functions for large datasets such as the mouse brain dataset[5] with 17,086 spots. We ran all experiments using one single graphics processing unit, and the timings of training processes for these ST datasets are presented in Supplementary Table 2.

Denoising and imputation of gene-expression levels

STitch3D's cell-type deconvolution results also enable denoising of expression levels of low-quality genes and imputation of unmeasured gene expressions in ST datasets. For the genes of interest to be denoised or imputed, STitch3D first calculates a cell-type-specific gene-expression profile vector $\mathbf{V}' = [V'_c] \in \mathbb{R}^{C\times 1}$ based on the scRNA-seq reference, in the same way as the cell-type-specific gene-expression profile matrix V of highly variable genes is obtained. Then, using STitch3D's estimated cell-type proportion in each spot β_i, the denoised or imputed expression levels of the genes of interest are given by $\sum_{c=1}^{C} \beta_{i,c} V'_c$.

ROC analysis and AUC scores for evaluating cell-type deconvolution accuracy in real data studies

We adopted AUC scores to assess the reliability of cell-type deconvolution results. Here we use the human DLPFC data analysis as an example to illustrate how AUC scores were calculated. The spatial study of DLPFC provided layer annotation on their generated DLPFC Visium slices. The single-cell dataset contains ten excitatory neuronal subtypes in cortical layers. The layer specificity of each neuronal subtype contained in the single-cell reference was studied[23]. For example, the neuronal subtype Ext_8_L5_6 was found expressing cortical layers 5–6 marker genes in the single-cell study, indicating its localization in layers 5–6. Based on this information, we evaluated the reliability of cell-type deconvolution methods by checking whether they successfully identified localization of subtype Ext_8_L5_6 in layers 5–6 by their estimated cell-type distributions. To this end, according to manual annotation on DLPFC Visium slices, we labelled spots in layers 5–6 as true and other spots as false, as the ground truth. We then adopted ROC analysis, and used the estimated cell-type proportions of the subtype Ext_8_L5_6 for a binary classification task to predict spot labels, resulting in an ROC curve. The AUC score calculates the area under the ROC curve, and can be used to indicate whether the estimated cell-type proportions are distributed in correct areas or not. We conducted the same ROC analyses for other neural subtypes. To make a fair comparison, the output of each method was normalized such that the sum of all cell-type proportions in each spot equalled one, before being used for evaluation of AUC scores.

Overall accuracy scores used in simulation studies

In simulation studies, we simulated ST data by gridding cells of high-resolution spatial datasets (with cell-type annotations) into spatial spots each containing multiple cells. Hence, cell-type compositions of spots in the simulated ST datasets are known. To make a fair comparison, the output of each cell-type deconvolution method was normalized such that the sum of all cell-type proportions in each spot equalled one. We used performance metrics, including the Pearson correlation coefficient (PCC), structural similarity index (SSIM)[60], root-mean-square error (r.m.s.e.) and Jensen-Shannon distance (JS), to quantitatively compare estimated cell-type proportions with the ground truth in each spot. The detailed description of these metrics is included in Supplementary Section 1. After calculating these metrics, four sets of ranks among the M methods were obtained, that is, $\mathrm{Rank}_{\mathrm{PCC}}$, $\mathrm{Rank}_{\mathrm{SSIM}}$, $\mathrm{Rank}_{\mathrm{rmse}}$ and $\mathrm{Rank}_{\mathrm{JS}}$. Each set of ranks ranges from

https://doi.org/10.1038/s42256-023-00734-1

1 to M, such that a higher rank value means a better performance in terms of the corresponding metric. The overall accuracy score of a method was computed by leveraging its ranks with respect to the four metrics:

$$\text{Accuracy, score} = \frac{1}{4M}(\text{Rank}_{\text{PCC}} + \text{Rank}_{\text{SSIM}} + \text{Rank}_{\text{rmse}} + \text{Rank}_{\text{JS}})$$

If the accuracy score of a method equals one, it means that this method outperforms the others in all four metrics.

Evaluation of the colocalization between cell types

We used the Pearson correlation coefficient between estimated cell-type proportions as an evaluation of the colocalization between cell types. Let $\beta_i = [\beta_{i,1}, \beta_{i,2}, \cdots, \beta_{i,C}]^T \in \mathbb{R}^C$ with $\beta_{i,c} \geq 0$, $\sum_{c=1}^{C} \beta_{i,c} = 1$ represent estimated cell-type proportions for spot i, where $i = 1, 2, \cdots, N$. The spatial colocalization between cell type u and cell type v is d efined as

$$R_{uv} = \frac{\sum_{i=1}^{N}\left(\beta_{i,u} - \sum_{j=1}^{N}\beta_{j,u}/N\right)\left(\beta_{i,v} - \sum_{j=1}^{N}\beta_{j,v}/N\right)}{\sqrt{\sum_{i=1}^{N}\left(\beta_{i,u} - \sum_{j=1}^{N}\beta_{j,u}/N\right)^2 \sum_{i=1}^{N}\left(\beta_{i,v} - \sum_{j=1}^{N}\beta_{j,v}/N\right)^2}}$$

Enrichment analysis of cell types

We calculated the enrichment score of a cell type in a region in the following way. First, the mean cell-type proportion of the region is computed and denoted as s_0. We then calculated the mean cell-type proportion of the region with shuffled spot indices 1,000 times, resulting in a collection of values, denoted $s_1, \ldots, s_{1,000}$. The enrichment score is then calculated as $(s_0 - \text{mean}(s_1, \ldots, s_{1,000}))/\text{std}(s_1, \ldots, s_{1,000})$. In the enrichment plot, positive enrichment scores are shown in red, and negative scores are shown in blue, and the absolute values are shown by varying dot sizes.

Data availability

All data used in this work are publicly available through online sources: human dorsolateral prefrontal cortex dataset profiled by Visium platform[6] (http://spatial.libd.org/spatialLIBD/); human dorsolateral prefrontal cortex dataset profiled by 10x Genomics Chromium platform[23] (GSE144136); mouse cortex dataset profiled by seqFISH+ (ref. 25) (https://github.com/CaiGroup/seqFISH-PLUS); mouse primary visual cortex dataset profiled by SMART-seq[26] (https://portal.brain-map.org/atlases-and-data/rnaseq/mouse-v1-and-alm-smart-seq); mouse visual cortex dataset profiled by STARmap[27] (https://kangaroo-goby.squarespace.com/data); mouse hypothalamic preoptic dataset profiled by MERFISH[28] (Dryad); mouse hypothalamic preoptic dataset profiled by Illumina NextSeq 500 (ref. 28) (GSE113576); mouse whole brain dataset profiled by ST platform[5] (GSE147747); mouse brain dataset profiled by 10x Genomics Chromium platform[16] (E-MTAB-11115); human embryonic heart dataset profiled by ST platform[7] (https://data.mendeley.com/datasets/dgnysc3zn5/1); human embryonic heart dataset profiled by 10x Genomics Chromium platform[7] (https://data.mendeley.com/datasets/mbvhhf8m62/2); murine lymph node spatial dataset profiled by Visium platform and scRNA-seq dataset profiled by 10x Genomics Chromium platform[18] (GSE173778); mouse skin sections profiled by Visium platform[34] (GSE178758); mouse skin dataset profiled by 10x Genomics Chromium platform[35] (GSE142471); HER2-positive breast tumour dataset profiled by ST platform[36] (https://doi.org/10.5281/zenodo.4751624); HER2-positive breast tumour dataset profiled by 10x Genomics Chromium platform[37] (GSE176078); *Drosophila* embryo dataset profiled by Stereo-seq[8] (https://db.cngb.org/stomics/datasets/STDS0000060); *Drosophila* embryo dataset profiled by sci-RNA-seq[39] (GSE190149); mouse olfactory bulb dataset profiled by Visium platform (https://www.10xgenomics.com/resources/datasets/adult-mouse-olfactory-bulb-1-standard-1); mouse olfactory bulb scRNA-seq dataset profiled by 10x Genomics Chromium platform[61] (GSE121891); mouse primary cortex 3D dataset profiled by STARmap[27] (https://kangaroo-goby.squarespace.com/data). Source data are provided with this paper.

Code availability

STitch3D software is available at https://github.com/YangLabHKUST/STitch3D. All codes are deposited in the Zenodo repository[62].

References

1. Ståhl, P. L. et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* **353**, 78–82 (2016).
2. Burgess, D. J. Spatial transcriptomics coming of age. *Nat. Rev. Genet.* **20**, 317–317 (2019).
3. Chen, W.-T. et al. Spatial transcriptomics and in situ sequencing to study Alzheimer's disease. *Cell* **182**, 976–991 (2020).
4. Moses, L. & Pachter, L. Museum of spatial transcriptomics. *Nat. Methods* **19**, 534–546 (2022).
5. Ortiz, C. et al. Molecular atlas of the adult mouse brain. *Sci. Adv.* **6**, eabb3446 (2020).
6. Maynard, K. R. et al. Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex. *Nat. Neurosci.* **24**, 425–436 (2021).
7. Asp, M. et al. A spatiotemporal organ-wide gene expression and cell atlas of the developing human heart. *Cell* **179**, 1647–1660 (2019).
8. Wang, M. et al. High-resolution 3D spatiotemporal transcriptomic maps of developing *Drosophila* embryos and larvae. *Dev. Cell* **57**, 1271–1283 (2022).
9. Briscoe, J. & Small, S. Morphogen rules: design principles of gradient-mediated embryo patterning. *Development* **142**, 3996–4009 (2015).
10. Lin, Y. & Yang, J. Y. 3D reconstruction of spatial expression. *Nat. Methods* **19**, 526–527 (2022).
11. Longo, S. K., Guo, M. G., Ji, A. L. & Khavari, P. A. Integrating single-cell and spatial transcriptomics to elucidate intercellular tissue dynamics. *Nat. Rev. Genet.* **22**, 627–644 (2021).
12. Zhao, E. et al. Spatial transcriptomics at subspot resolution with BayesSpace. *Nat. Biotechnol.* **39**, 1375–1384 (2021).
13. Hu, J. et al. SpaGCN: integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network. *Nat. Methods* **18**, 1342–1351 (2021).
14. Dong, K. & Zhang, S. Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nat. Commun.* **13**, 1739 (2022).
15. Cable, D. M. et al. Robust decomposition of cell type mixtures in spatial transcriptomics. *Nat. Biotechnol.* **40**, 517–526 (2022).
16. Kleshchevnikov, V. et al. Cell2location maps fine-grained cell types in spatial transcriptomics. *Nat. Biotechnol.* **40**, 661–671 (2022).
17. Biancalani, T. et al. Deep learning and alignment of spatially resolved single-cell transcriptomes with Tangram. *Nat. Methods* **18**, 1352–1362 (2021).
18. Lopez, R. et al. DestVI identifies continuums of cell types in spatial transcriptomics data. *Nat. Biotechnol.* **40**, 1360–1369 (2022).
19. Ma, Y. & Zhou, X. Spatially informed cell-type deconvolution for spatial transcriptomics. *Nat. Biotechnol.* **40**, 1349–1359 (2022).
20. Wolf, F. A. et al. PAGA: graph abstraction reconciles clustering with trajectory inference through a topology preserving map of single cells. *Genome Biol.* **20**, 59 (2019).

21. Haghverdi, L., Büttner, M., Wolf, F. A., Buettner, F. & Theis, F. J. Diffusion pseudotime robustly reconstructs lineage branching. *Nat. Methods* **13**, 845–848 (2016).
22. Gilmore, E. C. & Herrup, K. Cortical development: layers of complexity. *Curr. Biol.* **7**, R231–R234 (1997).
23. Nagy, C. et al. Single-nucleus transcriptomics of the prefrontal cortex in major depressive disorder implicates oligodendrocyte precursor cells and excitatory neurons. *Nat. Neurosci.* **23**, 771–781 (2020).
24. Li, B. et al. Benchmarking spatial and single-cell transcriptomics integration methods for transcript distribution prediction and cell type deconvolution. *Nat. Methods* **19**, 662–670 (2022).
25. Eng, C.-H. L. et al. Transcriptome-scale super-resolved imaging in tissues by RNA seqFISH+. *Nature* **568**, 235–239 (2019).
26. Tasic, B. et al. Shared and distinct transcriptomic cell types across neocortical areas. *Nature* **563**, 72–78 (2018).
27. Wang, X. et al. Three-dimensional intact-tissue sequencing of single-cell transcriptional states. *Science* **361**, eaat5691 (2018).
28. Moffitt, J. R. et al. Molecular, spatial and functional single-cell profiling of the hypothalamic preoptic region. *Science* **362**, eaau5324 (2018).
29. Oh, S. W. et al. A mesoscale connectome of the mouse brain. *Nature* **508**, 207–214 (2014).
30. Wang, Y. et al. Sprod for de-noising spatially resolved transcriptomics data based on position and image information. *Nat. Methods* **19**, 950–958 (2022).
31. Lein, E. S. et al. Genome-wide atlas of gene expression in the adult mouse brain. *Nature* **445**, 168–176 (2007).
32. Fang, Z., Liu, X. & Peltz, G. GSEApy: a comprehensive package for performing gene set enrichment analysis in Python. *Bioinformatics* **39**, btac757 (2022).
33. Evans, S. M., Yelon, D., Conlon, F. L. & Kirby, M. L. Myocardial lineage development. *Circ. Res.* **107**, 1428–1444 (2010).
34. Foster, D. S. et al. Integrated spatial multiomics reveals fibroblast fate during tissue repair. *Proc. Natl Acad. Sci. USA* **118**, e2110025118 (2021).
35. Haensel, D. et al. Defining epidermal basal cell states during skin homeostasis and wound healing using single-cell transcriptomics. *Cell Rep.* **30**, 3932–3947 (2020).
36. Andersson, A. et al. Spatial deconvolution of HER2-positive breast cancer delineates tumor-associated cell type interactions. *Nat. Commun.* **12**, 6012 (2021).
37. Wu, S. Z. et al. A single-cell and spatially resolved atlas of human breast cancers. *Nat. Genet.* **53**, 1334–1347 (2021).
38. Xu, D. et al. Endoplasmic reticulum stress targeted therapy for breast cancer. *Cell Commun. Signal.* **20**, 174 (2022).
39. Calderon, D. et al. The continuum of *Drosophila* embryonic development at single-cell resolution. *Science* **377**, eabn5800 (2022).
40. Tomancak, P. et al. Systematic determination of patterns of gene expression during *Drosophila* embryogenesis. *Genome Biol.* **3**, 1–14 (2002).
41. Tomancak, P. et al. Global analysis of patterns of gene expression during *Drosophila* embryogenesis. *Genome Biol.* **8**, 1–24 (2007).
42. Gramates, L. S. et al. FlyBase: a guided tour of highlighted features. *Genetics* **220**, iyac035 (2022).
43. Page, D. T. Inductive patterning of the embryonic brain in *Drosophila*. *Development* **129**, 2121–2128 (2002).
44. Crews, S. T. Drosophila embryonic CNS development: neurogenesis, gliogenesis, cell fate and differentiation. *Genetics* **213**, 1111–1144 (2019).
45. Hartenstein, V. *Atlas of Drosophila Development* (Cold Spring Harbor Laboratory Press, 1993).
46. Andrew, D. J., Henderson, K. D. & Seshaiah, P. Salivary gland development in *Drosophila melanogaster*. *Mech. Dev.* **92**, 5–17 (2000).
47. Maruyama, R. & Andrew, D. J. *Drosophila* as a model for epithelial tube formation. *Dev. Dyn.* **241**, 119–135 (2012).
48. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* **19**, 15 (2018).
49. Lécuyer, E. et al. Global analysis of mRNA localization reveals a prominent role in organizing cellular architecture and function. *Cell* **131**, 174–187 (2007).
50. Wilk, R., Hu, J., Blotsky, D. & Krause, H. M. Diverse and pervasive subcellular distributions for both coding and long noncoding RNAs. *Genes Dev.* **30**, 594–609 (2016).
51. Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902 (2019).
52. Korsunsky, I. et al. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat. Methods* **16**, 1289–1296 (2019).
53. Zhao, J. et al. Adversarial domain translation networks for integrating large-scale atlas-level single-cell datasets. *Nat. Comput. Sci.* **2**, 317–330 (2022).
54. Ming, J. et al. FIRM: flexible integration of single-cell RNA-sequencing data for large-scale multi-tissue cell atlas datasets. *Brief. Bioinform.* **23**, bbac167 (2022).
55. Vandereyken, K., Sifrim, A., Thienpont, B. & Voet, T. Methods and applications for single-cell and spatial multi-omics. *Nat. Rev. Genet.* **24**, 494–515 (2023).
56. Arun, K. S., Huang, T. S. & Blostein, S. D. Least-squares fitting of two 3-D point sets. *IEEE Trans. Pattern Anal. Mach. Intell.* **PAMI-9**, 698–700 (1987).
57. Zeira, R., Land, M., Strzalkowski, A. & Raphael, B. J. Alignment and integration of spatial transcriptomics data. *Nat. Methods* **19**, 567–575 (2022).
58. Blondel, V. D., Guillaume, J.-L., Lambiotte, R. & Lefebvre, E. Fast unfolding of communities in large networks. *J. Stat. Mech. Theory Exp.* **2008**, P10008 (2008).
59. Kingma, D. P. & Ba, J. Adam: a method for stochastic optimization. In *Proc. 3rd International Conference on Learning Representations (ICLR)* (ICLR, 2015).
60. Wang, Z., Bovik, A. C., Sheikh, H. R. & Simoncelli, E. P. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. Image Process.* **13**, 600–612 (2004).
61. Tepe, B. et al. Single-cell RNA-seq of mouse olfactory bulb reveals cellular heterogeneity and activity-dependent molecular census of adult-born neurons. *Cell Rep.* **25**, 2689–2703 (2018).
62. Wang, G. et al. STitch3D. *Zenodo* https://doi.org/10.5281/zenodo.8311065 (2023).
63. McInnes, L., Healy, J., Saul, N. & Grossberger, L. UMAP: Uniform Manifold Approximation and Projection. *J. Open Source Softw.* **3**, 861 (2018).
64. Fürth, D. et al. An interactive framework for whole-brain maps at cellular resolution. *Nat. Neurosci.* **21**, 139–149 (2018).
65. de Bakker, B. S. et al. An interactive three-dimensional digital atlas and quantitative database of human development. *Science* **354**, aag0053 (2016).

Acknowledgements

We acknowledge the following grants: Hong Kong Research Grant Council grants nos. 16301419, 16308120, 16307221 and 16307322, Hong Kong University of Science and Technology Startup Grants R9405 and Z0428 from the Big Data Institute, Guangdong-Hong Kong-Macao Joint Laboratory grant no. 2020B1212030001 and the RGC Collaborative Research Fund grant no. C6021-19EF to C.Y.; Hong Kong Research Grant Council grant no. 16209820, Lo Ka Chung Foundation through the Hong Kong Epigenomics Project, Chau Hoi Shuen Foundation, the SpatioTemporal Omics Consortium (STOC) and the STOmics Grant Program to A.R.W.; Hong Kong Research Grant Council grant no. 16103620, the Shenzhen Science and

Technology Innovation Commission JCYJ20180223181229868 and JCYJ20200109140201722 to Y.Y.

Author contributions

G.W., J.Z., A.R.W. and C.Y. conceived the idea. G.W. and J.Z. developed the method. A.R.W. and C.Y. supervised the project. G.W., J.Z., Y.Y., A.R.W. and C.Y. designed the experiments, performed the analyses and wrote the paper. Y.W. provided critical feedback during the study and helped revise the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42256-023-00734-1.

Correspondence and requests for materials should be addressed to Angela Ruohao Wu or Can Yang.

Peer review information *Nature Machine Intelligence* thanks Mengjie Chen, Miguel Esteban and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Mirko Pieropan, in collaboration with the *Nature Machine Intelligence* team.